

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

School of Information and Communications Technology

Software Requirement Specification

Version 1.3

AIMS – An Internet Media Store

Subject: ITSS Software Development

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1. Introduction

1.1. Objective

- The AIMS's SRS defines the e-commerce software's functional and non-functional requirements and acts as a guiding document.
- The purpose of this Software Requirements Specification is to provide a comprehensive, structured reference that defines what software must do, ensuring alignment between stakeholders, developers, and testers

1.2. Scope

- The AIMS is an e-commerce software that runs 24/7, and it is easy for new users to use. It can serve up to 1,000 customers simultaneously without significantly reducing performance and can operate continuously for 300 hours without failure. If something goes wrong, the software can get back to normal within 1 hour. The maximum response time of the software is 2 seconds under normal conditions or 5 seconds during peak hours.
- This software helps with online shopping, making it simple for users and businesses. It manages sales and customer needs. Its goal is to stay fast, dependable, and user-friendly, keeping customers happy and businesses running smoothly.

1.3. Glossary

<i>No</i>	<i>Term</i>	<i>Explanation</i>	<i>Example</i>	<i>Note</i>
1	VAT (Value-Added Tax)	A consumption tax placed on a product. In AIMS, the original value and current price of products are recorded without a 10% VAT.	The total product price including VAT is calculated and displayed on the invoice before payment.	Shipping fees are not subject to tax.

<i>No</i>	<i>Term</i>	<i>Explanation</i>	<i>Example</i>	<i>Note</i>
2	Sandbox	A testing environment used by developers to simulate operations without affecting a live system or processing real transactions.	PayPal Sandbox , VietQR Sandbox.	Used in this project for demonstration purposes only.
3	Administrator	A system role with privileges to manage user accounts. They can create, view, update, delete, block, unblock users, reset passwords, and assign roles.	An IT admin resetting a forgotten password for a product manager.	Requires login to access corresponding features.
4	Product Manager	A system role responsible for inventory and order management. They can add, view, edit, or delete products, manually adjust stock, and review pending orders.	A manager manually reducing stock due to damaged items.	Requires login to access corresponding features.
5	Pending Processing State	The status of an order immediately after a successful payment but before it is	An order paid via VietQR waiting for a manager to	Customers can choose to cancel the order while it is in this state.

<i>No</i>	<i>Term</i>	<i>Explanation</i>	<i>Example</i>	<i>Note</i>
		reviewed by a product manager.	approve it for shipping.	
6	VietQR	A standardized QR code payment method used in Vietnam. It allows customers to scan a generated code to complete their transaction.	Scanning the default QR code displayed on the checkout screen to pay.	Integrated for demonstration via the VietQR Test Callback API.
7	Deactivated Status	A product status indicating that an item is no longer available for sale. This occurs when a product manager attempts to delete a product that still has stock greater than zero.	A manager tries to delete a book with 5 copies left; the system marks it as "deactivated" instead.	Prevents deletion of items that still physically exist in the inventory.
8	Endpoint	A specific URL within an API used to execute a specific function or access a resource.	POST /v2/payments/captures/{capture_id}/refund.	Used in the PayPal REST API integration to issue refunds.
9	ISSN	International Standard Serial Number, an optional identifier	Entering an ISSN when adding a new	Part of the product metadata required

<i>No</i>	<i>Term</i>	<i>Explanation</i>	<i>Example</i>	<i>Note</i>
		used to record detailed information specifically for newspaper products.	newspaper to the catalog.	for specific media types.

1.4. References

2. Overall Description

2.1. Survey

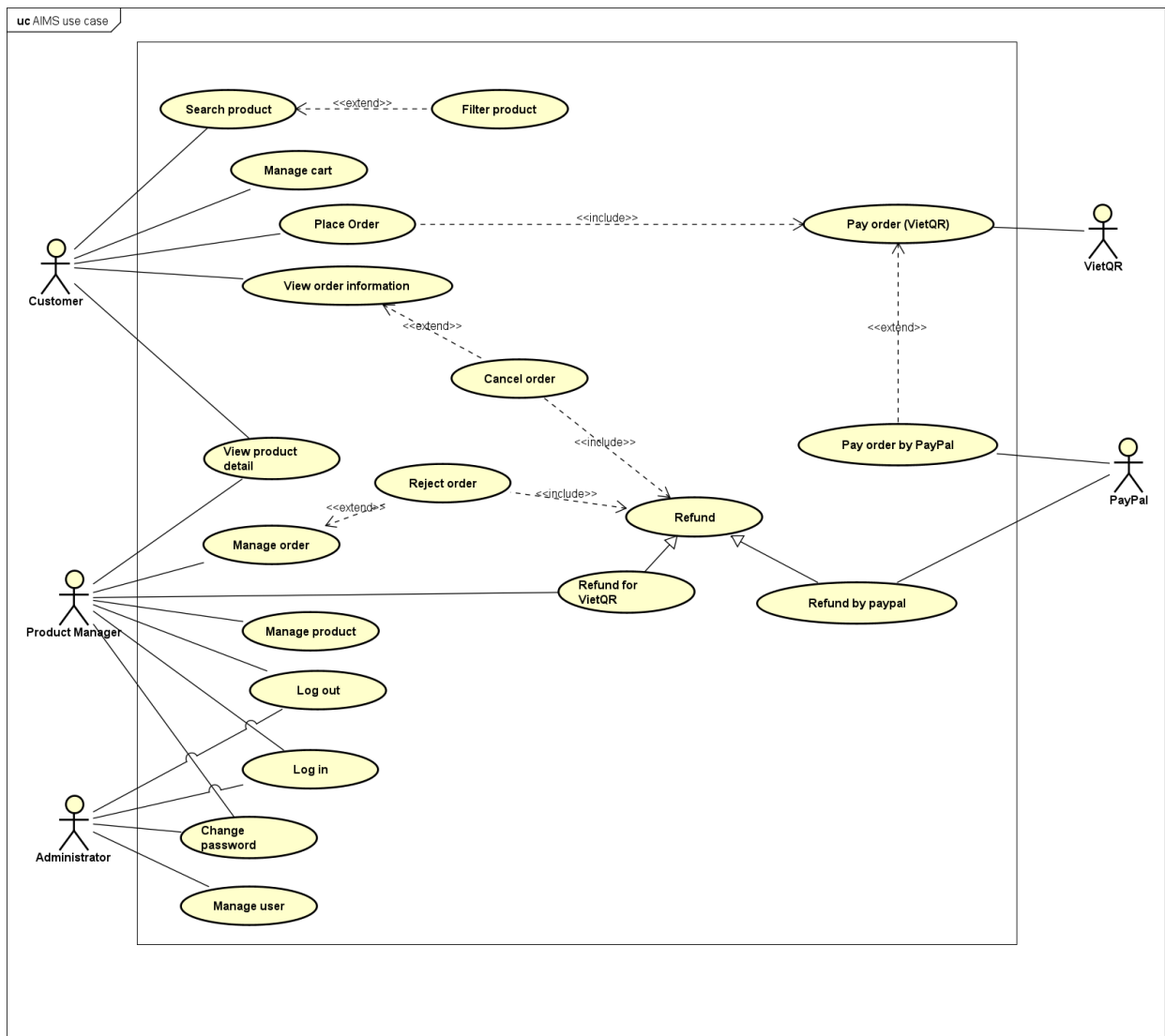
AIMS is an online media store system that supports both customer shopping activities and internal management operations. The system allows customers to search for products, view product details, add products to the cart, manage the cart, place orders, make payments, and track order information. At the same time, the system also provides management functions for Product Manager and Administrator to ensure that products, orders, and user accounts are handled effectively.

There are three main actors in the system. The first actor is Customer, who uses AIMS to browse and purchase media products such as books, CDs, DVDs, or LP records. The second actor is Product Manager, who is responsible for managing product information and checking customer orders. The third actor is Administrator, who manages user accounts and controls access permissions in the system.

Besides the main actors, AIMS also interacts with external services during the payment process. In this system, VietQR supports QR-based payment, while PayPal Sandbox is used for credit card payment simulation. These external services help the system complete payment-related functions in a more practical way.

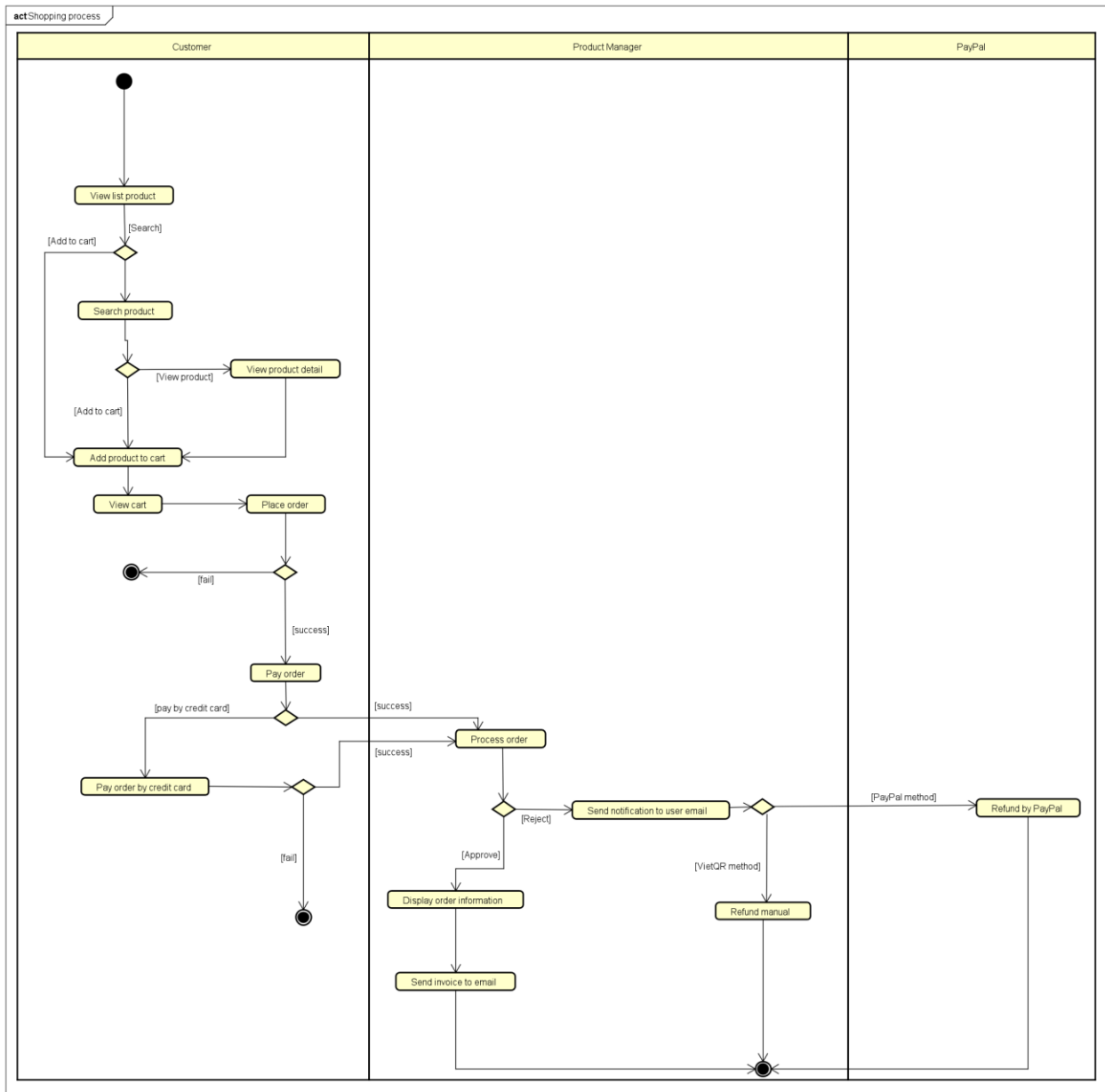
In general, AIMS is designed as an integrated system that supports the full process of online shopping and store management. Customers can interact with the store conveniently, while managers and administrators can perform their tasks through dedicated system functions.

2.2. Overall requirements

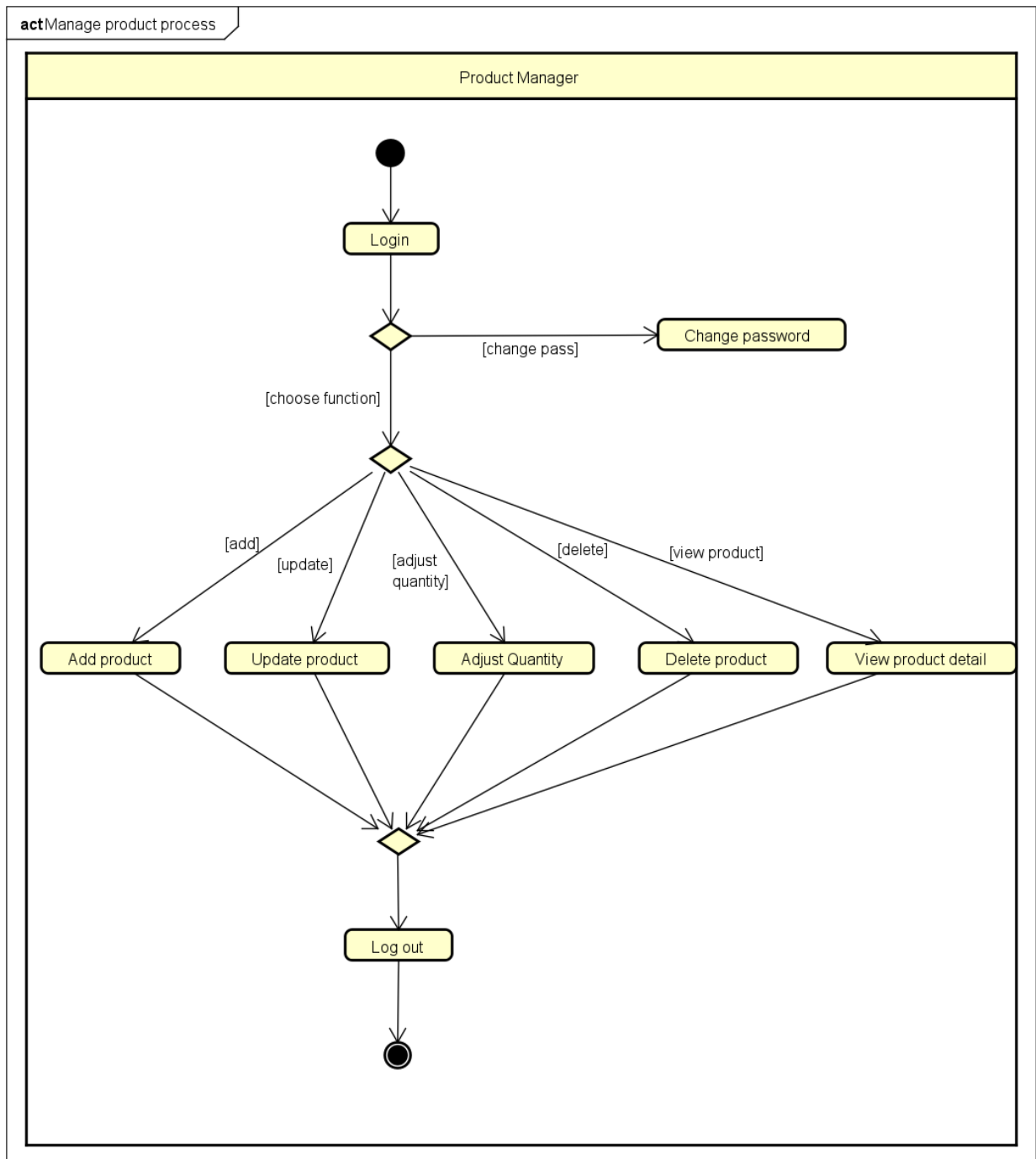


2.3. Business process

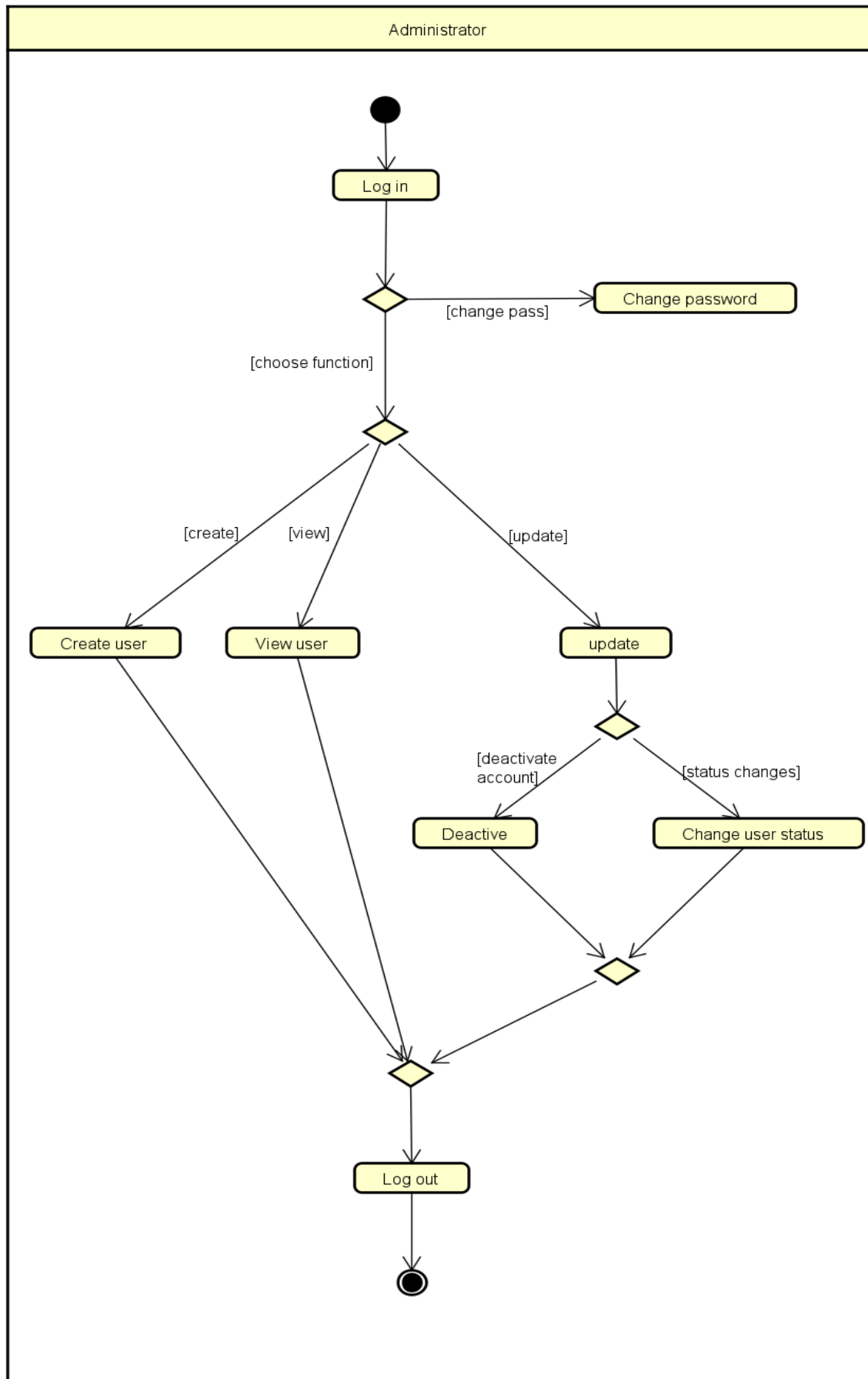
Shopping process



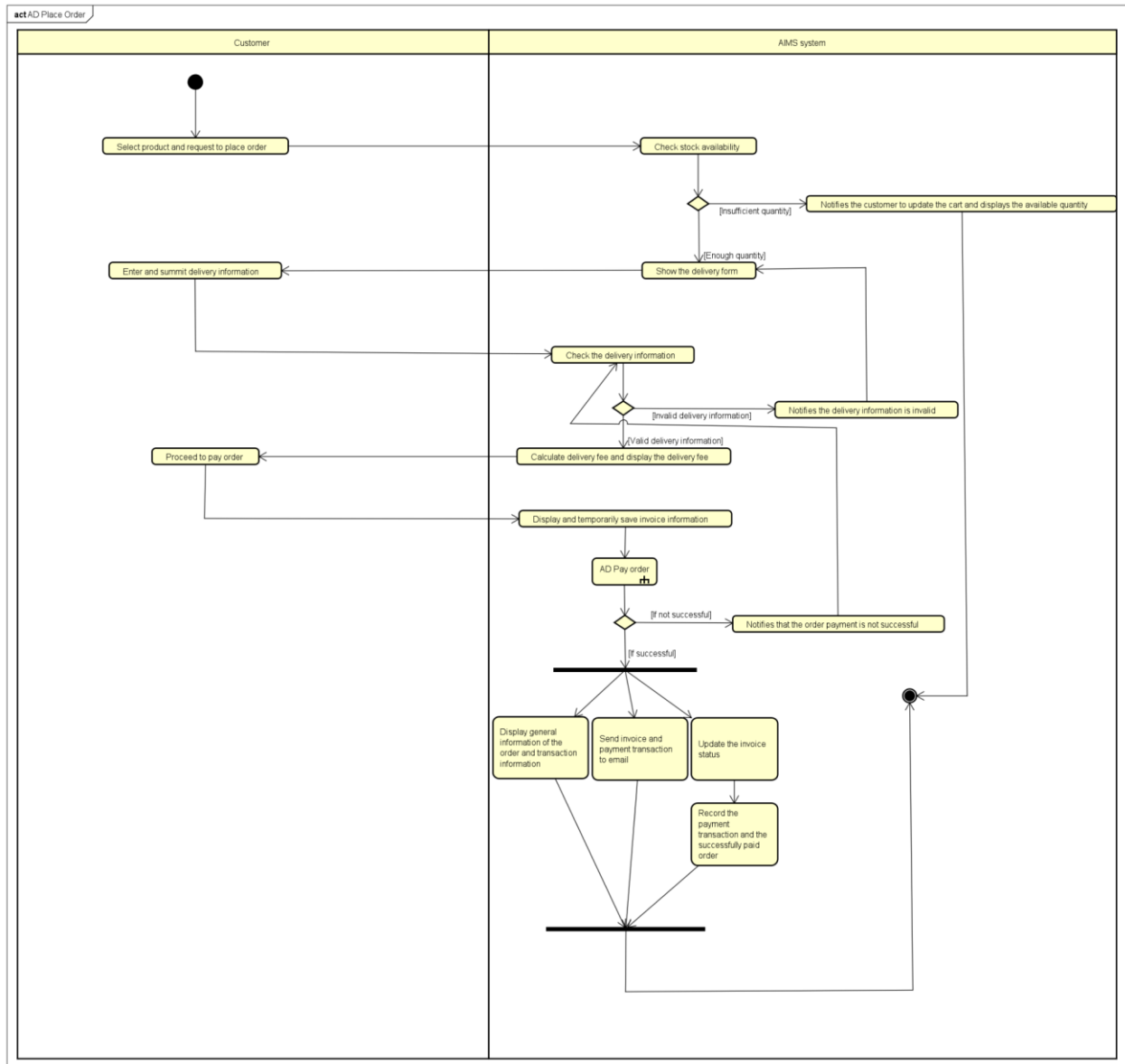
Manage product process



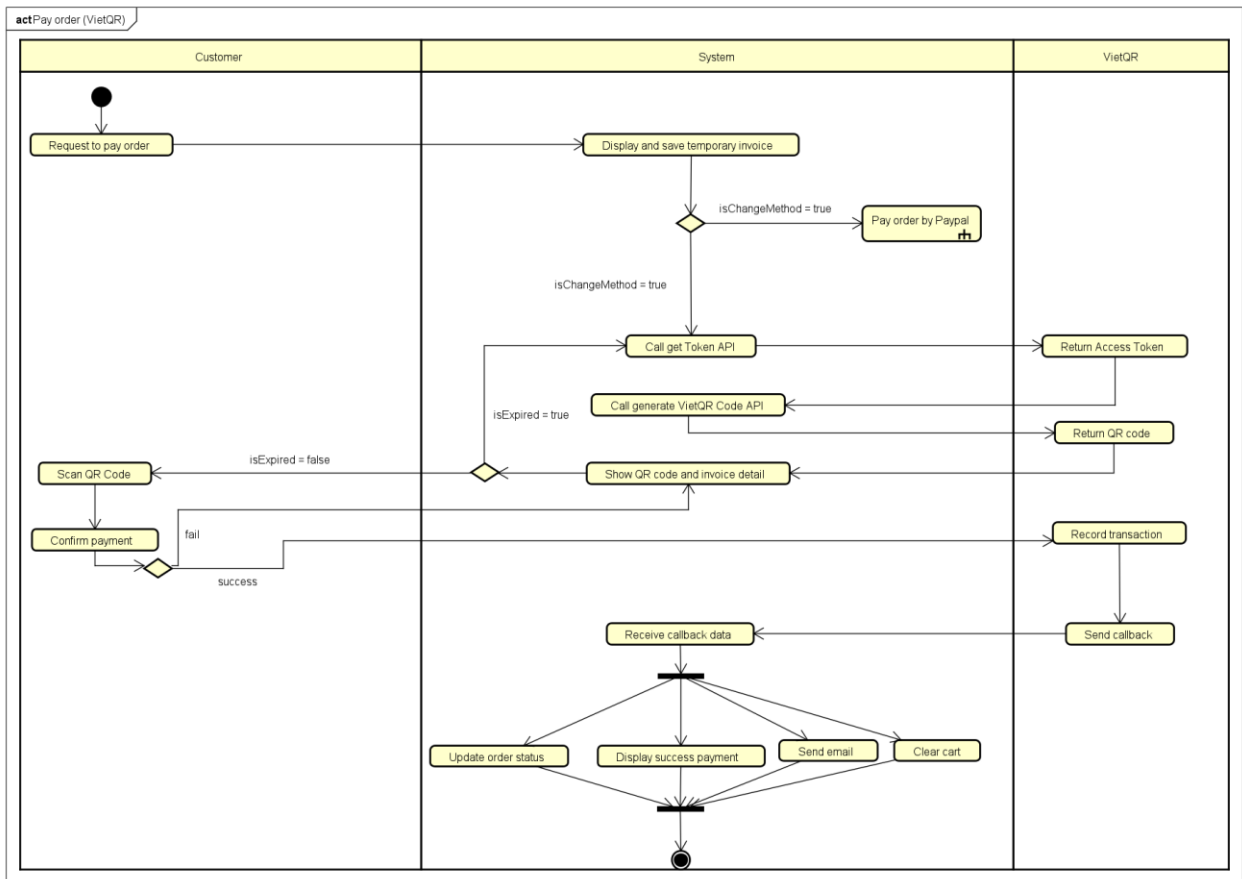
Manage user process



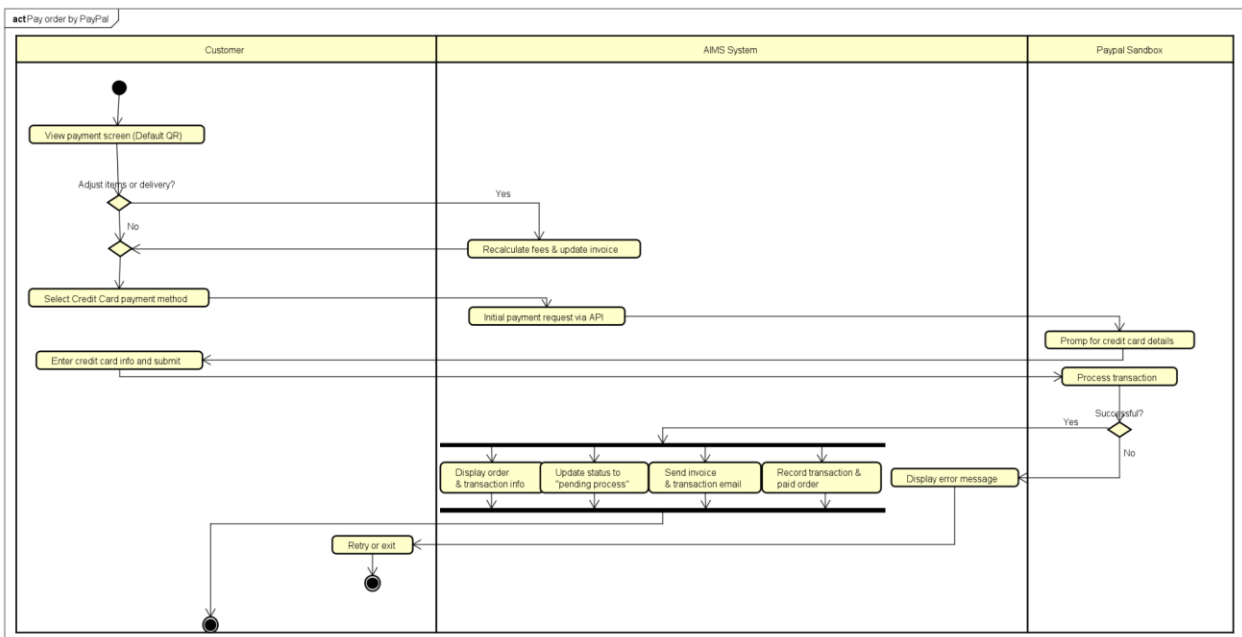
2.3.1. Activity Diagram “Place Order”



2.3.2. Activity Diagram “Pay Order (VietQR)”

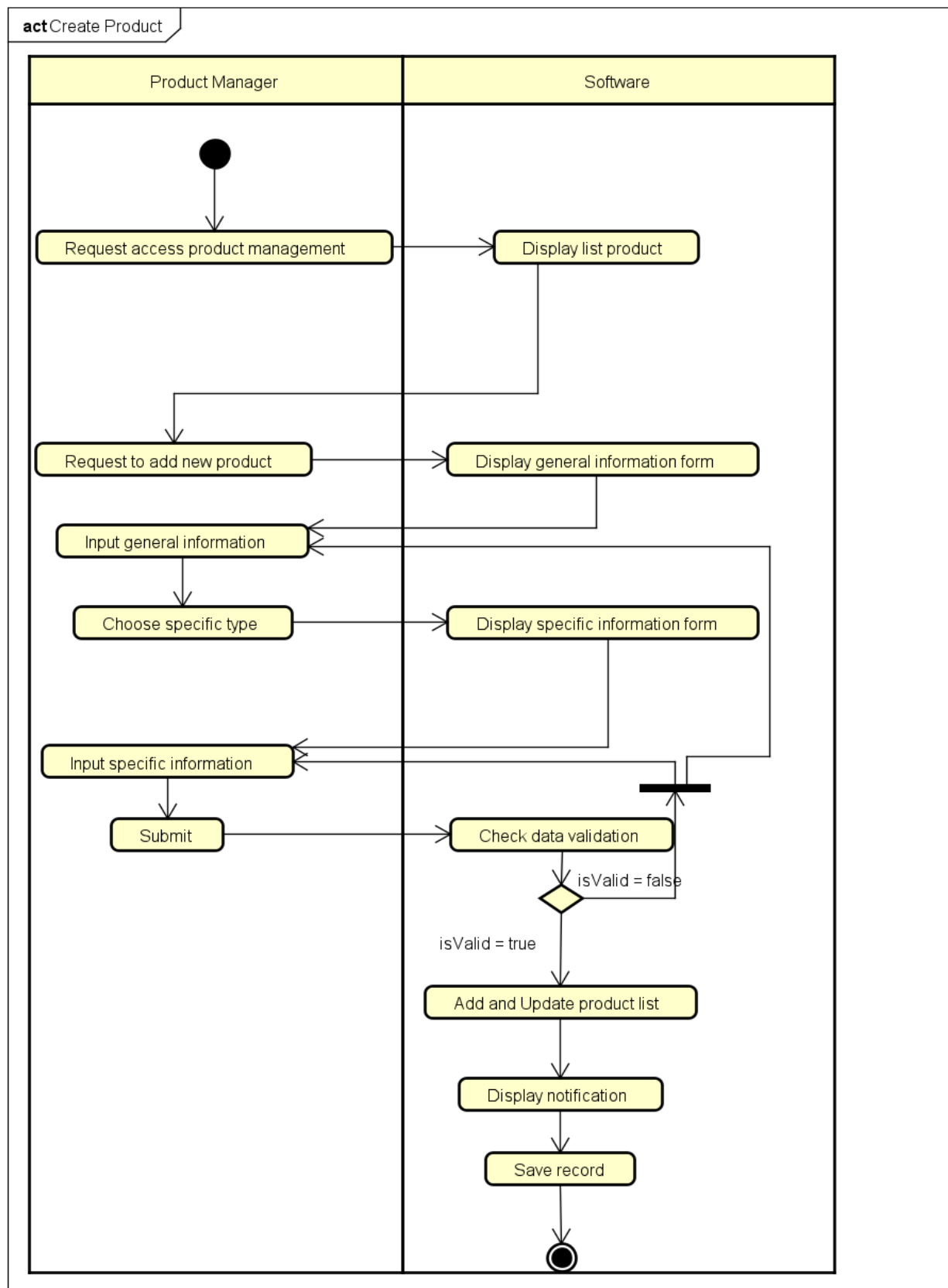


2.3.3. Activity Diagram “Pay Order by Credit Card”

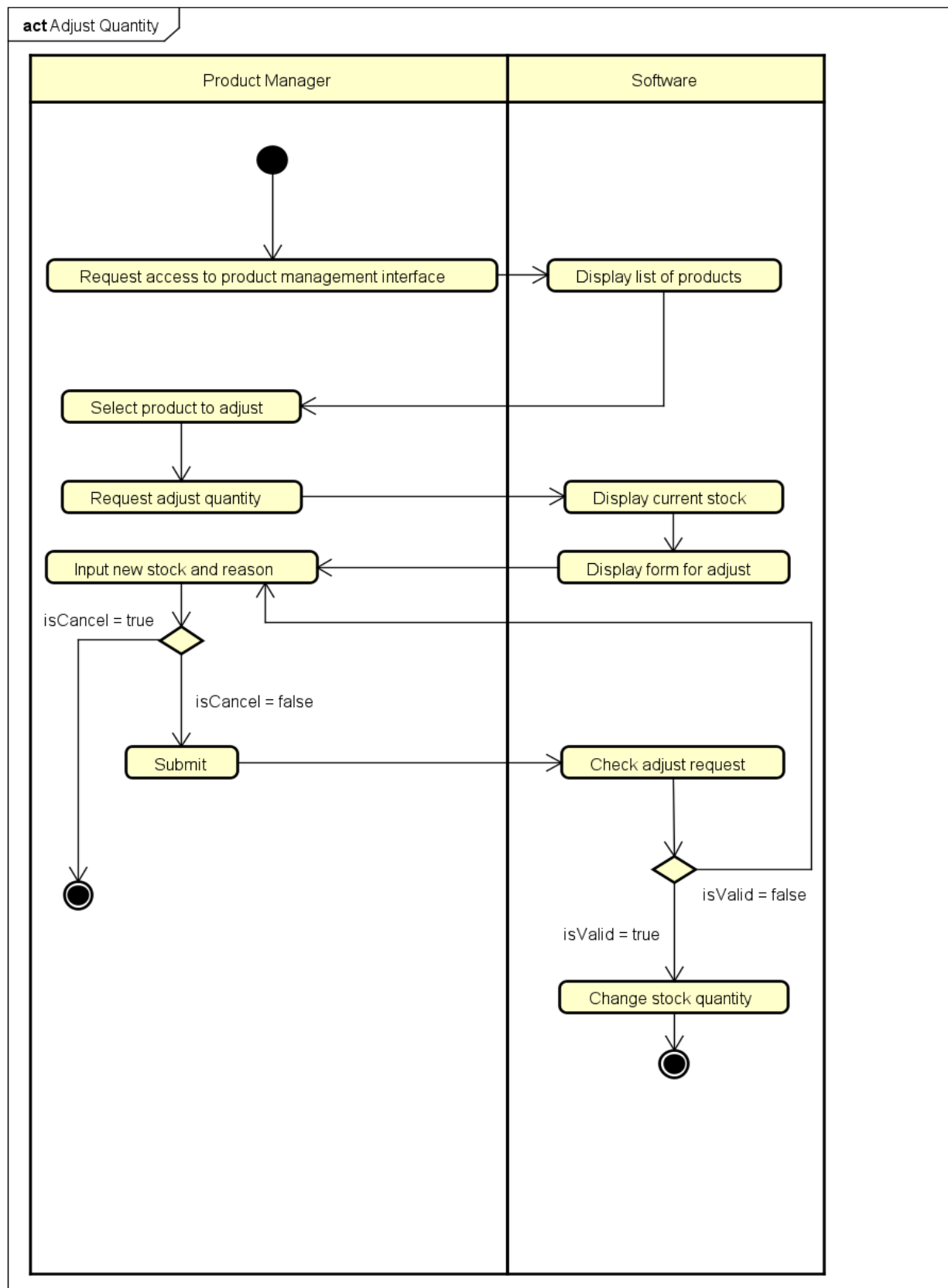


2.3.4. Activity Diagram “CUD Product”

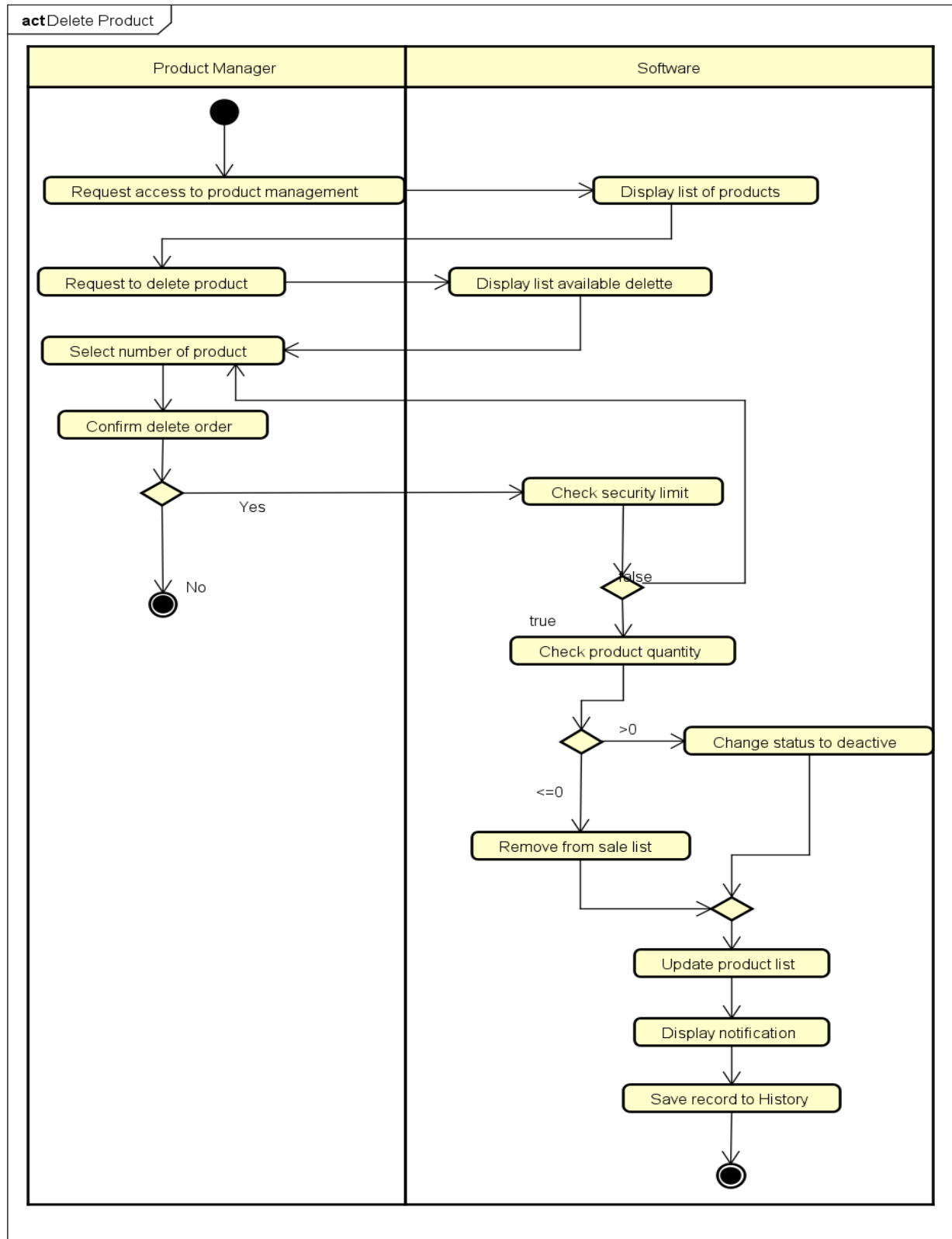
ADD PRODUCT



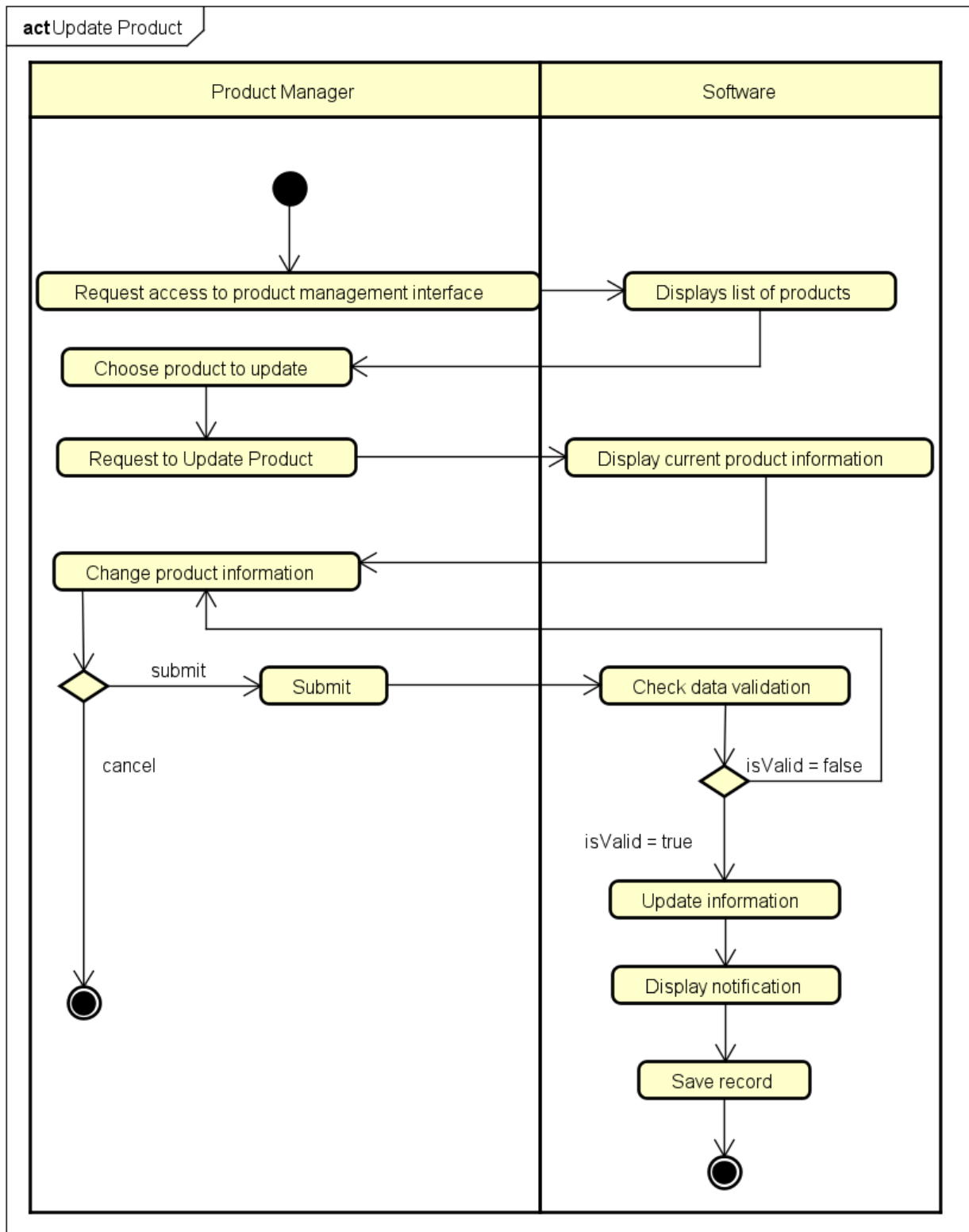
ADJUST QUANTITY



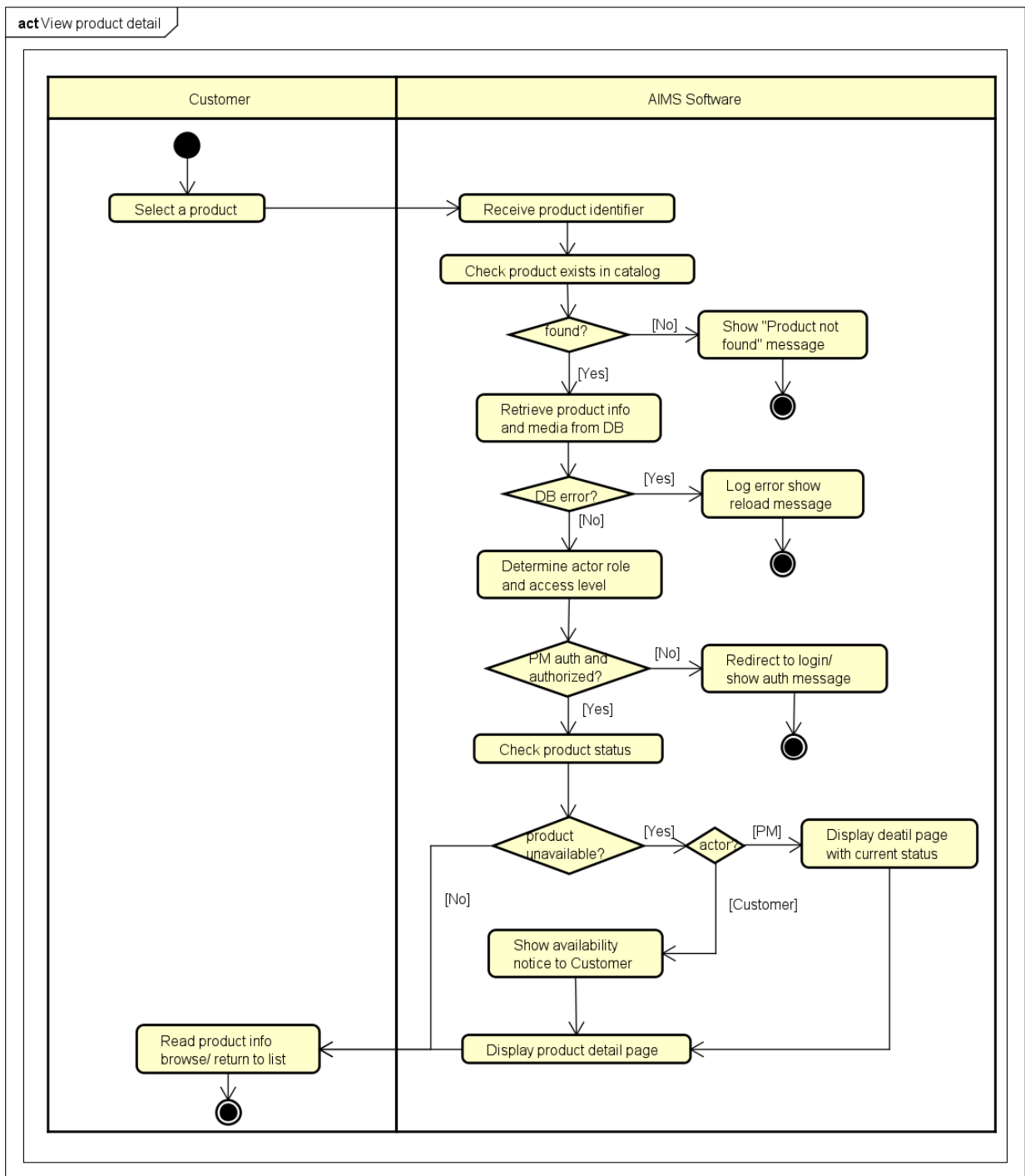
DELETE PRODUCT



UPDATE PRODUCT



2.3.5. Activity Diagram “View Product Detail”



3. Detailed Requirements

3.1. Use case “Place order”

Use Case “Place order”

1. Use case code

UC231

2. Brief Description

The use case “Place order” describes the interaction between customers and AIMS software

3. Actors

3.1 Customer

4. Preconditions

- The available stock is sufficient to fulfill the customer’s demand.
- There is at least one item in the cart.

5. Basic Flow of Events

1. Customer request to place order
2. The AIMS software checks the availability of products in the cart
3. AIMS software displays the form of delivery information with order information
4. Customer enters and submit delivery information (see Table 1)
5. AIMS software checks if the delivery information is valid
6. AIMS software calculates(or recalculates) shipping fee
7. Customer ask to pay order
8. AIMS software display and temporarily save invoice information (see Table 2)
9. AIMS software calls the UC “Pay order”
10. AIMS software will display general information of the order and transaction information(see Table 3)
11. AIMS software update the invoice status

12. AIMS software send invoice and payment transaction information to the customer's email

13. AIMS software records the payment transaction and the successfully paid order

6. Alternative flows

Table N-Alternative flows of events for UC Place order

No	Location	Condition	Action	Resume location
1.	At Step 2	If the available stock is insufficient	The AIMS software notifies the customer to update the cart and displays the available quantity for each affected product	Use case ends
2.	At Step 5	If the delivery info is invalid	AIMS software notifies that the delivery info is invalid (blank or wrong format)	At Step 3
3.	At Step 9	If the order payment is not successful or goes back from payment	AIMS software notifies that the order payment is not successful	At Step 5

7. Input data

Table 1-Input data of delivery information

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Receiver Name	Name of the recipient, used for delivery	Yes	Must contain only letters, not empty	Bui Trung Hieu

2.	Phone number	Recipient's phone number	Yes	- 10 digits -May contains separator “.” or space or “-” in the middle of the phone number(one separator type)	0981413168 0981 413 167 0988-097-136 0987-676-767
3.	Province	Province or city of the recipient	Yes	Must be selected from the provided list	Hanoi
4.	Address	Detailed street address of the recipient	Yes	Cannot be empty	No. 1 Dai Co Viet road, Bach Mai Ward, Hai Ba Trung District
5.	Shipping instructions	Additional instructions for delivery (optional)	No	Can be left empty; text up to 200 characters	Ship directly to the house

8. Output data

Table 2-Output data of invoice screen

No	Data fields	Description	Display format	Example
1.	Title	Title of a media product	Text, letters, and numbers	DVD Phim Sieu Nhan
2.	Price	Price of corresponding media product	- Comma for thousands separator - Positive integer - Right alignment	234,000
3.	Quantity	Quantity of the corresponding media	- Positive integer - Right alignment	3

4.	Amount	Total money of the corresponding media	- Comma for thousands separator - Positive integer - Right alignment	702,000
5.	Subtotal before VAT	Total price of products in the cart before VAT	- Comma for thousands separator - Positive integer - Right alignment	1,234,000
6.	Subtotal	Total price of all products in the cart with VAT	- Comma for thousands separator - Positive integer - Right alignment	1,357,400
7.	Shipping fee	Fee for delivering the order	- Comma for thousands separator - Positive integer - Right alignment	35,000
8.	Total	Sum of subtotal and shipping fee	- Comma for thousands separator - Positive integer - Right alignment	1,392,400

Table 3-Output data of general information of order and transaction info

No	Data fields	Description	Display format	Example
1.	Customer name	Name of the customer	Text, letters only, cannot be empty	Bui Trung Hieu
2.	Phone number	Customer's phone number	10 digits, may include spaces or dashes ("-"), one type only	0936363636
3.	Province	Province or city of the customer	Selected from a predefined list	Hanoi
4.	Address	Detailed street address of the customer	Text, cannot be empty	No. 1 Dai Co Viet road, Bach Mai Ward, Hai Ba

				Trung District
5.	Total amount	Total payment amount	-Right alignment -Vienamese currency(VND) -Vietnam locale	1,234,000
6.	Transaction ID	Unique identifier of the transaction	Number	123123123
7.	Transaction content	Description of the transaction	Text	Purchase of Media Product
8.	Transaction date	Date of the transaction	dd/mm/yy	16/03/2026

9. Postconditions

A new order is created, and its information is sent via email to the customer or nothing happens if payment is not successful

3.2. Use case “Pay Order(VietQR)”

Use Case Pay Order(VietQR)

1. Use case code

UC232

2. Brief Description

This use case describes the interaction between the customer, the AIMS software, and the VietQR system when the customer makes a payment for an order using a QR code. The process begins when the system displays invoice information along with the payment method and ends when the transaction is successfully recorded.

3. Actors

3.1 Customer

3.2 VietQR

4. Preconditions

- The Customer has completed entering shipping information and confirming the order (finishing the preparation steps).
- The system has calculated the shipping fee and generated temporary invoice.

5. Basic Flow of Events

1. Customer request to pay order
2. System display and save a temporary invoice information
3. System call VietQR's Get Token API to obtain an Access Token to access the system
4. VietQR return access token to System
5. System call Generate VietQR Code API use token to create QR code for payment base on total money of invoice
6. VietQR return QR code can be image or data string to System.
7. System show the QR code and detail invoice.
8. Customer scan QR code
9. Customer confirm the payment
10. VietQR record successful transactions send notification to System.
11. System receive callback data and recheck order ID and amount
12. System update status order to pending and store detail transaction
13. System display successful payment with order information and transaction code
14. System clear cart
15. System send email confirm invoice and transaction information to Customer's email

6. Alternative flows

Table 1 - Alternative flows of events for UC "Pay Order By Credit Card"

No	Location	Condition	Action	Resume location
4	At step 2	If customer want to change payment method	System change to usecase "Pay Order by Credit Card".	Use case end

5.	At any step	If the customer want to change order information.	System call “Place Order” use case	Use case end
6.	At step 9	Transaction is failed	System notify the failure and allow customer try scanning again	At step 7
7.	At step 10	If Network timeout / No callback received	Customer clicks "I have paid". System calls "Check Transaction" API to verify.	At step 11
8.	At Step 11	If QR code expired	System notify expired QR code, request to refresh	At step 3

7. Input data

Table 2 - Input data of UC “Pay order by credit card”

No	Data fields	Description	Mandatory	Valid condition	Example
6.	Access Token	Token for API authentication	Yes	Validated by VietQR	eyJhbGci..
7.	Amount	Total money to pay	Yes	Positive integer	1,200,000
8.	Order ID	Unique ID for reconciliation	Yes	Matches existing order	ORDER_123
9.	Callback Data	Notification from VietQR	Yes	Valid signature/checksum	JSON Object

8. Output data

Table 3 - Output data of UC “Pay order by credit card”

No	Data fields	Description	Display format	Example
10.	QR Code	Visual code for scanning	Image/String	
11.	Total amount	Final amount paid	Vietnamese currency (VNĐ)	1.200.000 VNĐ
12.	Transaction ID	Provided by VietQR	String	VQR123456
13.	Transaction date	Date of successful payment	dd/mm/yyyy	16/03/2026

9. Postconditions

- A new order is created with a "Pending" status.
- System sent the invoice and payment transaction information to the customer's email.
- System save recorded of payment transaction information and order detail in the history.
- System clear Customer's cart.

3.3. Use case “Pay Order By Credit Card”

Use Case “Pay Order By Credit Card”

1. Use case code

- UC233

2. Brief Description

This use case describes the process of a customer paying for an order using a credit card as an alternative to the default QR code payment method. The credit card transaction is processed through the integrated PayPal Sandbox environment via PayPal REST APIs v2. Upon successful payment, the system displays the transaction and order details, sends a confirmation email containing the invoice and transaction information, records the payment and order data, and updates the order status to pending processing.

3. Actors

3.1 Primary Actor: Customer

3.2 Secondary Actor: PayPal SandBox

4. Preconditions

- The customer has set up delivery information.
- The software displays and temporarily saves invoice information, including the list of products in the cart, quantity, product prices, total product price excluding VAT, total product price including VAT, delivery fee, and total amount to be paid.
- The software initially displays a QR code as the default payment method.
- The AIMS system is configured to connect to the PayPal Sandbox environment.

5. Basic Flow of Events

1. The customer chooses to switch from the default QR code payment method to the alternative payment method using a credit card.
2. The AIMS system initiates a payment request to the integrated PayPal Sandbox environment using PayPal REST APIs v2.
3. The PayPal Sandbox prompts the customer for their credit card information.
4. The customer enters their credit card details and submits the payment.
5. The PayPal Sandbox captures the payment and returns a successful transaction confirmation to the AIMS system.
6. After a successful payment, the software displays general information of the order (customer name, phone number, shipping address, province, total amount) and transaction information (transaction ID, transaction content and transaction datetime).
7. The software sets the order to a pending processing state.
8. The software sends invoice and payment transaction information to the customer's email.
9. The software records the payment transaction information and the successfully paid order.

6. Alternative flows

Table 1 - Alternative flows of events for UC “Pay Order By Credit Card”

No	Location	Condition	Action	Resume location
1	At Step 1	If the customer decides to adjust the delivery method or the items they wish to purchase.	The software recalculates the delivery fees and updates the corresponding invoice.	Resumes at Step 1

2	At any step during the payment process	If the customer chooses to go back to any previous step or exit the software.	The software halts the payment process and navigates the user back to the previous screen or exits.	Use case ends
3	At Step 5	If the PayPal Sandbox fails to capture the payment (e.g., declined card or invalid API credentials).	The software displays an error message and does not proceed with the successful post-payment actions.	Resumes at Step 3 or Use case ends

7. Input data

Table 2 - Input data of UC “Pay order by credit card”

No	Data fields	Description	Mandatory	Valid condition	Example
1	Payment Method Selection	The choice to switch from the default QR code to a credit card.	Yes	Must be selected as "Credit Card" (PayPal Sandbox).	"Credit Card" (PayPal Sandbox). Credit Card
2	Credit Card Information	The payment credentials required by the PayPal Sandbox integration to capture the payment.	Yes	Must be valid test credentials accepted by the PayPal Sandbox environment.	Sandbox Credit Card Number, Expiry Date, CVV

3	Payment Payload	The total amount and order details sent from AIMS to PayPal Sandbox API.	Yes	Must match the total amount to be paid.	USD 12.50
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8. Output data

Table 3 - Output data of UC “Pay order by credit card”

No	Data fields	Description	Display format	Example
1.	General Order Information	Displays the customer name, phone number, shipping address, province, and total amount.	Text	Nguyen Van A, 0987654321, Co Viet, Hanoi, 125,000 VND
2.	Transaction Information	Displays the transaction ID, transaction content, and transaction datetime.	Text	TXN-98765, Payment for #123, 2023-10-27 14:30:00
3.	Email Notification	Invoice and payment transaction information sent to the customer's email.	Email format	(Email containing the invoice transaction details)

9. Postconditions

- The order status is updated to a pending processing state.
- The software has sent the invoice and payment transaction information to the customer's email.
- The software has recorded the payment transaction information and the successfully paid order in the system.

3.4. CUD Product

3.4.1. Add Product

Use Case “Add Product”

1. Use case code

UC234.1

2. Brief Description

This use case allows the Product Manager to add a new media product (Book, Newspaper, CD, DVD) to the AIMS system. The system will check data format constraints and pricing rules before officially saving it to the database.

3. Actors

3.1 Product Manager

4. Preconditions

- Product Manager successfully logged into the AIMS System with appropriate account provided.
- The AIMS System is currently operating normally..

5. Basic Flow of Events

1. Product Manager request access to the product management interface.
2. Software displays list of products and options for adding, editing, deleting and adjusting quantities.
3. Product Manager select Add Product
4. Software displays general form for Product Manager
5. Product Manager input general information and choose specific type of product.
6. Software displays form for fill specific information depending on product type
7. Product Manager input specific information and submit.
8. Software checks the data and price rule.
9. Software update product list
10. Software display notifications and save record into history

6. Alternative flows

Table 1 - Alternative flows of events for UC “Add Product”

No	Location	Condition	Action	Resume location
1.	At step 8	If the input data is not in the correct format	Software display detailed error for each data field	At step 3 or 5
2.	Any step	If Product Manager cancel process	Software do not save any changes and return product list interface	End the use case

7. Input data**Table 2 - Input data of UC “Add Product”**

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Barcode	Unique identifier for the product	Yes	Unique in the system; numeric only	893123456789
2.	Title	The name of the media product	Yes	Cannot be blank	DVD Hello World
3.	Category	Product classification	Yes	One of: Books, Newspapers, CDs, DVDs	Books
4.	General Description	Condition, primary color, or return condition	Yes	Text description	New condition, blue cover

5.	Original Value	Base price excluding 10% VAT	Yes	Positive integer	100
6.	Current Price	Selling price excluding 10% VAT	Yes	Between 30% and 150% of Original Value	120
7.	Dimensions	Physical size (Height x Width x Length)	Yes	Positive values only	20 x 15 x 3 (cm)
8.	Weight	Physical weight of the product	Yes	Positive value (kg)	0.5
9.	Author	Creator(s) of the book	Yes (Book)	Required if Category is "Books"	Ngo Tat To
10.	Cover Type	Format of the book cover	Yes (Book)	"Paperback" or "Hardcover"	Paperback
11.	Editor-in-chief	Head of the newspaper	Yes (News)	Required if Category is "Newspapers"	Nguyen Van A
12.	Artists	Performers or singers	Yes (CD)	Required if Category is "CDs"	Son Tung M-TP
13.	Track List	List of songs with titles and lengths	Yes (CD)	Must include title and duration per track	Track 1 - 10:30
14.	Disc Type	Physical format of the disc	Yes (DVD)	"Blu-ray" or "HD-DVD"	Blu-ray

15.	Director	Director of the media	Yes (DVD)	Required if Category is "DVDs"	Christopher Nolan
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8. Output data

Table 3 - Output data of UC “Add Product”

No	Data fields	Description	Display format	Example
1.	Operation Status	Notification of the result for the performed action (add, edit, delete, or adjust).	Text notification	Product added successfully
2.	Error Message	Detailed notification when input data is invalid or violates system rules.	Text (Red)	Price must be between 30% and 150% of Original Value
3.	History Log	A record list of all product additions, edits, and deletions.	Timeline/Log entry	16/03/2026: Manager A added product XXXX
4.	Product List	A list of all media products currently available in the store's inventory.	Data table	(List of Books, CDs, ...)

9. Postconditions

- Interface Refresh: The product management interface is automatically refreshed to display the most recent data after the operation.

- Audit trail: A new record created in history, log the detail information of the operation
- User notification: Product Manager receive notification from software, product list interface must refreshed to display updated list
- Stored data: All products successfully saved into database or new information be updated with newest version.
- Error message: Product Manager receive error text message at invalid field.

3.4.2. Update Product

Use Case “Update Product”

1. Use case code

UC234.2

2. Brief Description

This use case allows the Product Manager to update information for a single media product in the AIMS system at a time. It ensures the current price remains between 30% and 150% of the original value, with both figures excluding 10% VAT. Depending on the category, the system validates specific mandatory fields such as authors, publishers, or directors. The software records all changes in the history log and notifies the user of any invalid data or rule violations.

3. Actors

3.1 Product Manager

4. Preconditions

- Product Manager successfully logged into the AIMS System with appropriate account provided.
- The AIMS System is currently operating normally.

5. Basic Flow of Events

1. Product Manager request access to the product management interface.
2. Software displays list of products and options for adding, editing, deleting and adjusting quantities.
3. Product Manager select one product
4. Product Manager select Edit Function
5. Software displays product's current information
6. Product Manager update the necessary information
7. Product Manager submit the form
8. Software check constraint of data and price
9. Software update product information
10. Software display notifications and save record into history.

6. Alternative flows

Table 1 - Alternative flows of events for UC "Update Product"

No	Location	Condition	Action	Resume location
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1.	At step 8	If the input data is not in the correct format	Software display detailed error for each data field	At step 6
2.	Any step	If Product Manager cancel process	Software do not save any changes and return product list interface	At step 3

7. Input data

Table 2 - Input data of UC “Update Product”

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Title	The name of the media product	Yes	Cannot be blank	DVD Hello World
2.	Category	Product classification	Yes	One of: Books, Newspapers, CDs, DVDs	Books
3.	General Description	Condition, primary color, or return condition	Yes	Text description	New condition, blue cover
4.	Original Value	Base price excluding 10% VAT	Yes	Positive integer	100
5.	Current Price	Selling price excluding 10% VAT	Yes	Between 30% and 150% of Original Value	120
6.	Dimensions	Physical size (Height x	Yes	Positive values only	20 x 15 x 3 (cm)

		Width x Length)			
7.	Weight	Physical weight of the product	Yes	Positive value (kg)	0.5
8.	Author	Creator(s) of the book	Yes (Book)	Required if Category is "Books"	Ngo Tat To
9.	Cover Type	Format of the book cover	Yes (Book)	"Paperback" or "Hardcover"	Paperback
10.	Editor-in-chief	Head of the newspaper	Yes (News)	Required if Category is "Newspapers"	Nguyen Van A
11.	Artists	Performers or singers	Yes (CD)	Required if Category is "CDs"	Son Tung M-TP
12.	Track List	List of songs with titles and lengths	Yes (CD)	Must include title and duration per track	Track 1 - 10:30
13.	Disc Type	Physical format of the disc	Yes (DVD)	"Blu-ray" or "HD-DVD"	Blu-ray
14.	Director	Director of the media	Yes (DVD)	Required if Category is "DVDs"	Christopher Nolan

8. Output data

Table 3 - Output data of UC “Update Product”

No	Data fields	Description	Display format	Example
1.	Operation Status	Notification of the result for the	Text notification	Product updated successfully

		performed action (add, edit, delete, or adjust).		
2.	Error Message	Detailed notification when input data is invalid or violates system rules.	Text (Red)	Price must be between 30% and 150% of Original Value
3.	History Log	A record list of all product additions, edits, and deletions.	Timeline/Log entry	16/03/2026: Manager A updated information of Barcode X

9. Postconditions

- Data Persistence: Product information is accurately updated to reflect the latest version in the database.
- Error message: Product Manager receive error text message at invalid field.
- Interface Refresh: The product management interface is automatically refreshed to display the most recent data after the operation.
- Audit trail: A new record created in history, log the detail information of the operation
- User notification: Product Manager receive notification from software, product list interface must refreshed to display updated list

3.4.3. Adjust Quantity

Use Case “Adjust Quantity”

1. Use case code

UC234.3

2. Brief Description

This use case allows Product Manager to manually adjust the inventory quantity of a product (add to inventory or withdraw due to damage or loss). Any quantity changes

must be accompanied by a specific reason for auditing and inventory management purposes.

3. Actors

3.1 Product Manager

4. Preconditions

- Product Manager successfully logged into the AIMS System with appropriate account provided.
- The AIMS System is currently operating normally.

5. Basic Flow of Events

1. Product Manager request access to the product management interface.
2. Software displays list of products and options for adding, editing, deleting and adjusting quantities.
3. Product Manager select product to adjust
4. Product Manager select adjust quantity
5. Software display box of current stock and form for adjust
6. Product Manager input new stock and reason
7. Software check reason
8. Software change the stock quantity
9. Software update product list
10. Software display notifications and save record into history

6. Alternative flows

Table 1 - Alternative flows of events for UC “Adjust Quantity”

No	Location	Condition	Action	Resume location
1.	At step 7	If Product Manager do not input Reason field	▪ Software require to enter reason for adjust product quantity	At step 7
2.	Any step	If Product Manager cancel process	▪ Software do not save any changes and	End usecase

			return product list interface	
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7. Input data

Table 2 - Input data of UC “Adjust Quantity”

No	Data fields	Description	Mandatory	Valid condition	Example
1.	New Stock	Updated manual inventory count	Yes (Adjust)	Non-negative integer	50
2.	Reason	Explicit record for manual stock change	Yes (Adjust)	Required for all manual adjustments	Damaged during transit

8. Output data

Table 3 - Output data of UC “Adjust Quantity”

No	Data fields	Description	Display format	Example
1.	Stock Quantity	The actual number of items in stock after a manual update or a sale.	Positive integer	45
2.	Adjustment Reason	An explicit record explaining why a manual stock change was performed.	Text description	Damaged during transit
3.	Operation Status	Notification of the result for the performed action (add, edit, delete, or adjust).	Text notification	Product adjusted successfully

4.	History Log	A record list of all product additions, edits, and deletions.	Timeline/Log entry	16/03/2026: Manager A updated stock of Barcode X
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9. Postconditions

- Inventory Accuracy: The stock quantity of product is accurately updated in system according to manual adjustment
- Stock reason logging: Explicit reason is saved and linked to stock change record
- Audit trail: A new record created in history, log the detail information of the operation
- User notification: Product Manager receive notification from software, product list interface must refreshed to display updated list

3.4.4. Delete Product

Use Case “Delete Product”

1. Use case code

UC234.4

2. Brief Description

This use case lets the Product Manager remove products from the AIMS sales list. The system checks security rules, such as the maximum number of products that can be deleted, and verifies the stock quantity to decide whether the product will be completely deleted or only marked as inactive.

3. Actors

3.1 Product Manager

4. Preconditions

- Product Manager successfully logged into the AIMS System with appropriate account provided.
- The AIMS System is currently operating normally.

5. Basic Flow of Events

1. Product Manager request access to the product management interface.

2. Software displays list of products and options for adding, editing, deleting and adjusting quantities.
3. Product Manger select Delete Product
4. Product Manager select up to ten products to delete at a time.
5. Product Manager confirm the delete order
6. Software check security limit delete.
7. Software check product quantity
8. Software remove from sale list
9. Software update product list
10. Software display notifications and save record into history

6. Alternative flows

Table 1 - Alternative flows of events for UC “Delete Product”

No	Location	Condition	Action	Resume location
1.	At step 5	If Product Manager selects more than 10 products to delete at once.	Software displaying the warning only allows a maximum of 10 products to be selected.	At step 4
2.	At step 6	If total number of products that Product Manager deleted today over 20 products.	Software notify the Product Manager of a security violation and refused to delete	At step 4
3.	At step 7	If quantity of product removed greater than 0	Software automatic change status to deactive.	At step 9

7. Input data

Table 2 - Input data of UC “Delete Product”

No	Data fields	Description	Mandatory	Valid condition	Example
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1.	Selected Barcode	Unique identifier for the product	Yes	Barcode available in system	893123456789
2.	Confirmation	Notification that Product Manager confirm to delete	Yes	Boolean	Yes

8. Output data

Table 3 - Output data of UC “Delete Product”

No	Data fields	Description	Display format	Example
1.	Operation Status	Notification of the result for the performed action (add, edit, delete, or adjust).	Text notification	Products deleted successfully
2.	Product Status	The commercial availability of the product (available or deactivated).	Label/Tag	Deactivated
3.	Error Message	Detailed notification when input data is invalid or violates system rules.	Text (Red)	Can’t delete more product.
4.	History Log	A record list of all product additions, edits, and deletions.	Timeline/Log entry	16/03/2026: Manager A deleted(list product)
5.	Product List	A list of all media products currently available in the store's inventory.	Data table	(List of Books, CDs, ...)

9. Postconditions

- Product Status: Products with Stock = 0 are removed from the sales list; products with Stock > 0 have their status updated to "Deactivated".
- Security Update: The system updates the Product Manager's daily counter for the number of products deleted.
- Interface Refresh: The product management interface is automatically refreshed to display the most recent data after the operation.
- Audit trail: A new record created in history, log the detail information of the operation
- User notification: Product Manager receive notification from software, product list interface must refreshed to display updated list

3.5. Use case “View Product Detail”

Use Case View Product Detail (Customer / Product Manager)

1. Use case code

- UC235

2. Brief Description

- This use case describes the interaction between Customer / Product Manager and AIMS software when they want to view the detailed information of a selected product. After a product is chosen, the system retrieves its information, checks access rights when necessary, verifies the current product status, and displays the corresponding general and type-specific details on screen.

3. Actors

- Customer
- Product Manager

4. Preconditions

- The selected product exists in the system catalog.
- For Customer: the customer is on the product list screen, homepage, or search result screen.
- For Product Manager: the Product Manager is logged in and is on the product management screen.
- The product belongs to one of the supported product types in AIMS: Book, Newspaper, CD, or DVD.

5. Basic Flow of Events

1. The actor selects a product from the homepage, product list, search result, or management screen.
2. AIMS receives the request and identifies the selected product.
3. The system checks whether the selected product exists in the product catalog.
4. The system retrieves the selected product's information and related media from the database.
5. The system determines the actor's role and access level.
6. The system checks the current status of the selected product.
7. The system displays the product detail page with the information available to that actor.
8. The actor reads the product information and may continue browsing, return to the previous page, or choose another product.

6. Alternative Flows

No	Location	Condition	Action	Resume location
1	At step 3	If the selected product does not exist in the catalog	AIMS shows a "Product not found" message and allows the actor to return or continue browsing	Use case ends
2	At step 4	If a database or service error occurs while retrieving product information or media	AIMS logs the error and shows a reload / try again later message	Use case ends
3	At step 5	If the actor is a Product Manager but is not authenticated or authorized	AIMS redirects the actor to login or shows an authorization message	Use case ends

4	At step 6	If the selected product is unavailable for sale	For Customer, AIMS shows an availability notice before displaying the detail page. For Product Manager, AIMS displays the detail page with the current product status	Resume at Step 7
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7. Input Data

8. Output Data

Table B.1 – Output general data of product

No	Attribute	Meaning	Format	Sample value
1.	Product name	The official name displayed as the main title of the product page	Text	Doraemon Volume 1
2.	Product type	The category to which the product belongs	Text	Book
3.	Selling price	The current selling price shown to users	Positive number, VND	85,000
4.	Original value	The original value of the item for management purposes	Positive number, VND	70,000
5.	Barcode	Unique code used to identify the product	Non-empty alphanumeric string	BK10293847
6.	Description	A short introduction or summary of the product	Text paragraph	A popular children's comic book
7.	Weight	The physical weight of the product	Number with unit	350 g
8.	Dimensions	The size of the	Height × Width ×	20 × 14 × 2 cm

		product in standard order	Length	
9.	Image	Visual representation of the product	Image file / thumbnail	doraemon_vol1.png
10.	Stock quantity	Number of units currently available in inventory	Non-negative integer	18

Table B.2. Specific information for Book

No	Book attribute	Meaning	Example
1	Author	Person who wrote the book	Fujiko F. Fujio
2	Cover style	Physical cover format of the book	Paperback
3	Publisher	Organization that publishes the book	Kim Dong Publishing House
4	Publication date	Date when the book was published	15/01/2024
5	Number of pages	Total number of pages in the book	192
6	Language	Language used in the content	Vietnamese
7	Genre	Literary type or subject category	Comic

Table B.3. Specific information for Newspaper

No	Newspaper attribute	Meaning	Example
1	Editor-in-chief	Person responsible for editorial	Tran Trung Nghia

		management	
2	Publisher	Organization issuing the newspaper	Tuoi Tre News
3	Publication date	Date of issue	23/03/2026
4	Issue number	The edition number of the newspaper	No. 369
5	Release frequency	How often the newspaper is published	Daily
6	ISSN	International identification number for the newspaper	2734-9123
7	Language	Main language of the newspaper	Vietnamese
8	Main sections	Key content areas covered in the issue	Education,Sports

Table B.4. Specific information for CD

No	CD attribute	Meaning	Example
1	Artist / Band	Performer or group featured on the CD	Westlife
2	Record label	Company releasing the CD	Sony Music
3	Track list	Songs included in the album	1. My Love – 3:52
4	Music genre	General style of music	Pop
5	Release date	Official release date of the CD	9/9/2025

Table B.5. Specific information for DVD

No	DVD attribute	Meaning	Example
1	Disc format	Technical format of the disc	DVD-9
2	Director	Director of the movie or video content	Hayao Miyazaki
3	Duration	Total running time	169 minutes
4	Studio	Production or distribution studio	Studio Ghibli
5	Audio language	Main spoken language on the disc	Japanese
6	Subtitle options	Subtitle languages available	English, Vietnamese
7	Release date	Official release date	14/07/2023
8	Film genre	Category of the movie content	Animation/ Fantasy

9. Postconditions:

- The system shows the detailed information of the selected product if the request is valid.
- The actor can review the product information for the next actions such as continuing browsing, searching, or making management decisions.
- If an error occurs, the system displays an appropriate message and records the error log.
- No business data is changed after the use case is completed.

4. Supplementary specification

The supplementary specification provides additional requirements for the SRS document. These requirements include system functionality rules, usability, reliability, performance, and other constraints to ensure the system operates effectively.

4.1. *Functionality*

- The software should validate all user input data, including product information, user data, and shipping information, to ensure correct formatting and that all required fields are filled.
- The system should store a history of product additions, edits, and deletions performed by product managers.
- The system should automatically update product stock quantities whenever products are sold.
- The system should implement price constraints, whereby the current selling price of the product should be between 30% and 150% of the product's original value.
- The system should notify the product manager when an operation is invalid, such as entering incorrect data formats or violating product management rules.
- The system should maintain order information, payment transactions, and invoice data.

4.2. *Usability*

- The AIMS software should provide a clear and intuitive interface that allows customers to browse and purchase products without logging in.
- The system should allow customers to easily search for products by title or category and filter by price range.
- The system interface should allow users to view product details and manage shopping carts easily.
- Administrative functions are only accessible to users with administrator or product manager roles, and these users must log in to access their features.

4.3. *Reliability*

- The system should operate 24/7 and support up to 1,000 simultaneous customers without significantly reducing performance.
- The system should be able to run continuously for at least 300 hours without failure.
- In case of a system failure, the system should be able to recover and return to normal operation within 1 hour.

4.4. *Performance*

- The system's response time should not exceed 2 seconds under normal conditions.
- During peak hours, the maximum response time should not exceed 5 seconds.
- The system should efficiently handle shopping cart updates, order placement, and payment confirmation.

- The system should update inventory levels whenever products are sold.

4.5. *Supportability*

- The system should store the history of product additions, edits, and deletions.
- Software architecture should allow for future expansion, for example, to support additional product types or payment methods.

4.6. *Other requirements*

- The system must integrate payment services using PayPal Sandbox (credit card payments) and VietQR (QR code payments).
- The system should automatically generate and display QR codes for QR-based payments.
- The system must calculate dynamic shipping fees based on product weight and delivery location.
- Orders with a total product value exceeding 100,000 VND will receive free shipping up to a maximum of 25,000 VND.